SciFig-Evaluation: A Multilingual Benchmark and Behavioural Framework for Evaluating Multimodal Language Models on Scientific Figures

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Declaration

No portion of the work contained in this document has been submitted in support of an application for a degree or qualification of this or any other university or other institution of learning. All verbatim extracts have been distinguished by quotation marks, and all sources of information have been specifically acknowledged.

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Abstract

Multimodal large language models are increasingly asked to read scientific figures, yet existing benchmarks largely measure whether a model can extract a value from a chart, not whether it is a faithful visual reader. This thesis introduces SciFig-Eval, a multilingual benchmark of 1,005 figures drawn from 158 arXiv papers and paired with 1,411 expert annotations in English, Bulgarian, German and Chinese, together with a three-axis behavioural framework — Admittance, Resistance and Inductance — that separates honesty in the absence of evidence, faithfulness under misleading context, and sound inductive reasoning. Using an MQM-style multi-judge evaluation pipeline, the framework is exercised on eleven contemporary models spanning five families via five adversarial probe types. The resulting behavioural profile reveals systematic, near-orthogonal failure modes that conventional accuracy scores obscure.

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Contents

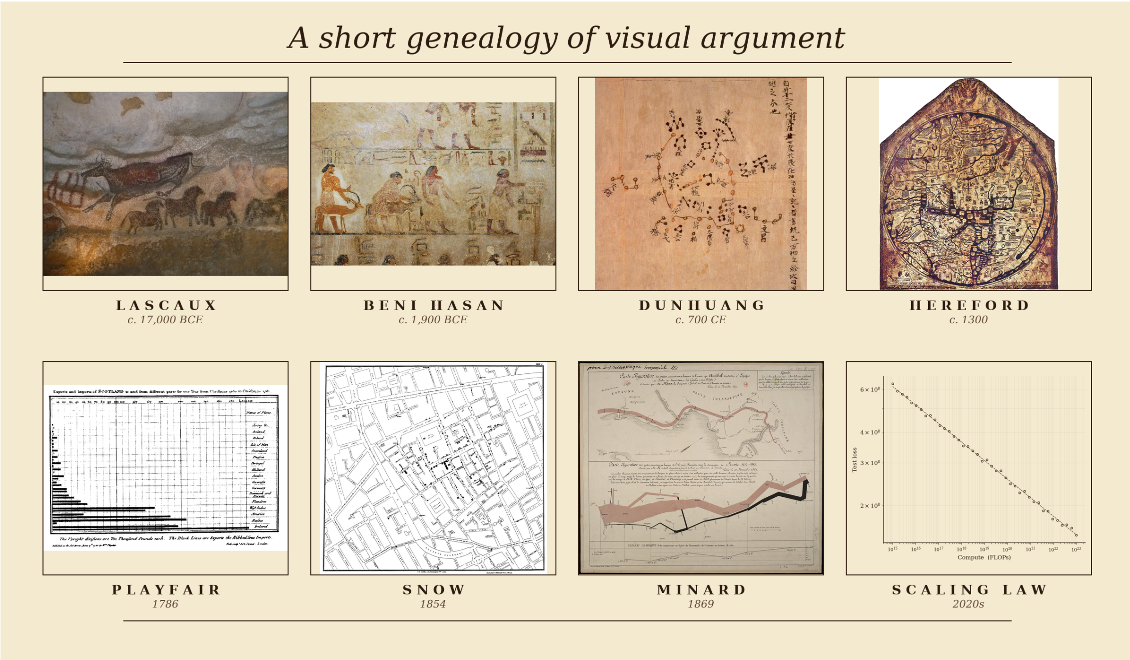
Chapter 1

Introduction

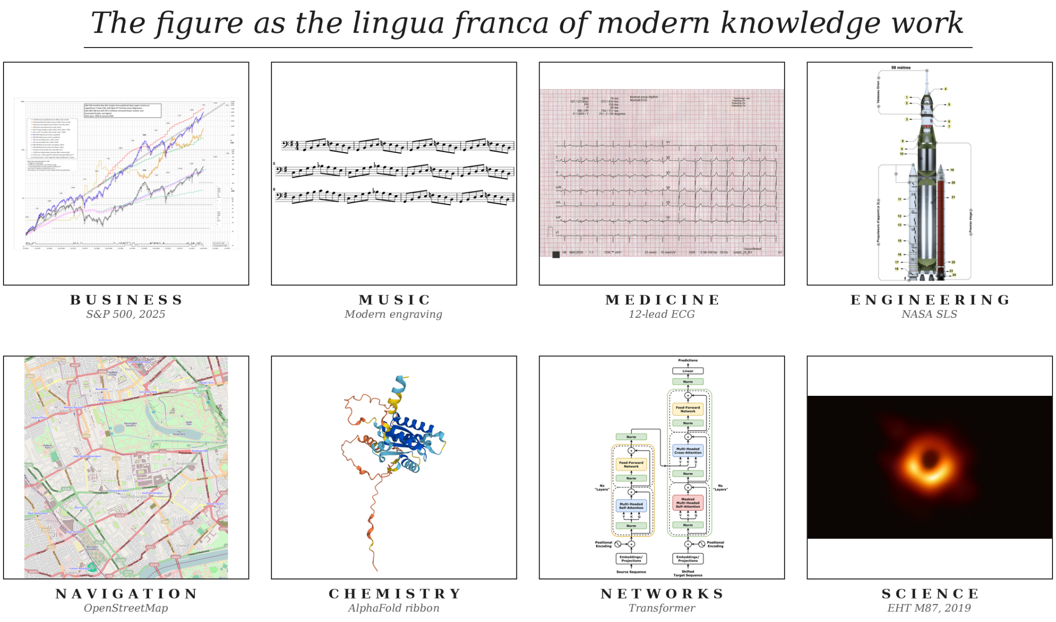
1.1 *Homo sapiens*: A Species of Visual Thinkers

The record of a species willing to reason pictorially begins no later than the Upper Palaeolithic. Some time between seventeen and fifteen millennia before the present, an inhabitant of what is now the Dordogne crushed haematite between stones, lit a pine torch, and rendered an aurochs on Lascaux limestone at approximately life size (Aujoulat, 2005). These visual arguments have continued to evolve. Twelfth-dynasty Egyptian painters at Beni Hasan committed a caravan of Semitic travellers, traditionally associated with the Joseph narrative, to a tomb wall around 1900 BCE (Kamrin, 2009); seventh-century Chinese astronomers committed over fourteen hundred stars to silk at Dunhuang (Bonnet-Bidaud et al., 2009); medieval cartographers compressed theology, geography and chronology onto the Hereford *mappa mundi* (Harvey, 1996). In 1786 William Playfair invented the bar chart in order to argue, graphically and in print, about Britain's trade balances (Playfair, 1786). John Snow's 1854 cholera map of Soho laid down the method of modern epidemiology (Johnson, 2006), and Charles Joseph Minard's 1869 flow map of Napoleon's 1812 Russian campaign (Friendly, 2002) is what Tufte (2001) has called, without serious contest, the densest argument ever made on paper (Figure 1.1a). When a human being intends a claim of any real quantitative weight, the first move is almost invariably to draw.

In contemporary practice, figures have not only evolved but have become significantly embedded in daily operations. Businesses read through dashboards of candlestick charts and cohort curves (Nison, 2001) that senior management consults in direct preference to prose; engineers coordinate the hundreds of thousands of components that constitute a ship, a hospital or an aircraft through hierarchies of blueprints read without ambiguity by people who may never speak to the designer (Ferguson, 1992); and even life-and-death decisions are made by reading a heartbeat off the grid of an electrocardiogram, or tissue contrast off a radiograph or perfusion map (Goldberger et al., 2017). Music (Treitler, 1984), satellite navigation (Kaplan and Hegarty, 2017), chemistry (Hoffmann, 1995) and network science (Newman, 2010) sustain the same diagrammatic primacy, and every competent empirical paper in the natural and social sciences eventually commits its argument to a figure (Figure 1.1b). **The figure has become a lingua franca of modern knowledge work**, and in many domains the argumentative weight it now carries is strictly greater than the weight carried by the accompanying prose.



(a)



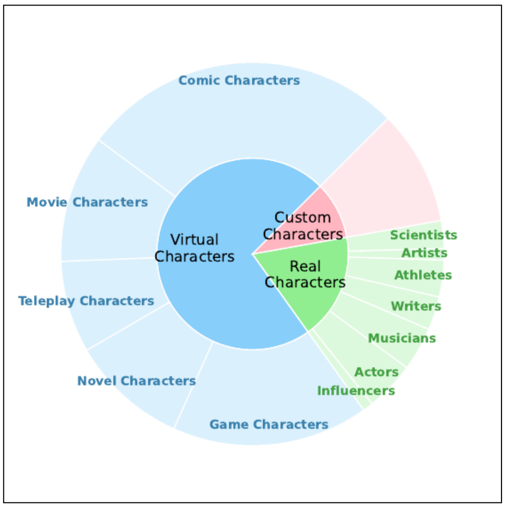
(b)

Figure 1.1: **(a)** A short genealogy of visual argument, from Lascaux and Egyptian wall painting through the medieval *mappa mundi*, Playfair's 1786 bar chart and Snow's 1854 cholera map to modern neural scaling curves. **(b)** The figure as a lingua franca of modern knowledge work across business, music, medicine, engineering, navigation, chemistry, networks and the natural sciences. *Sources.* **(a)** Lascaux aurochs, Beni Hasan caravan, Dunhuang star chart, Hereford *mappa mundi*, Playfair 1786 bar chart, Snow 1854 cholera map and Minard 1869 flow map of Napoleon's Russian campaign, all public domain (via Wikimedia Commons, the Metropolitan Museum of Art, the British Library, Hereford Cathedral, the Internet Archive and the Bibliothèque nationale de France respectively); scaling curves adapted from Kaplan et al. 2020. **(b)** A candlestick chart, score excerpt, ECG trace, blueprint, nautical chart, molecular diagram, network graph and scientific plot, from public-domain or openly-licensed repositories (Wikimedia Commons, NOAA, PhysioNet) or rendered by the author.

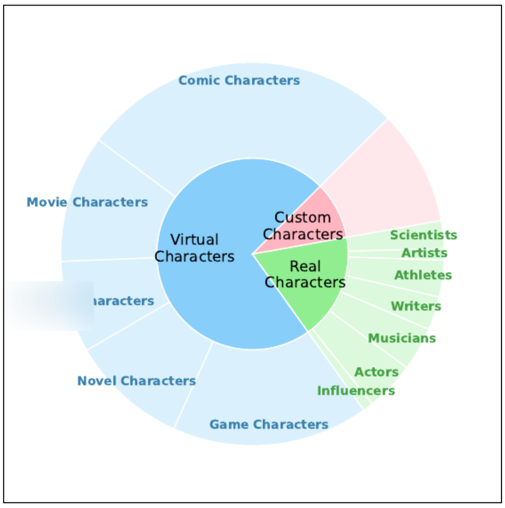
1.2 Can *Intelligentia Machinae* (Machine Intelligence) Comprehend Visuals Similarly?

Recent progress in artificial intelligence has rapidly normalised the idea that these systems can be integrated into nearly every part of working life. Having established that the human is a visual thinker and that the figure is a lingua franca of modern knowledge work, it becomes imperative to investigate how well the current frontier of artificial intelligence, in particular the vision-language models given their present dominance and, looking ahead, world models now emerging alongside them, can navigate the visual paradigm, since in collaborative work with humans they will more often than not encounter images.

There is evidence that the best models of today still fail in some way. Figure 1.2 reproduces a sunburst drawn from an arXiv paper in the SCIFIG-EVAL corpus. In the original image the outer-ring category between *Movie Characters* and *Novel Characters* is legibly *Teleplay Characters*; in the degraded variant, that single label has been selectively blurred. Asked which category sits between the two, GPT-5.2, arguably the most widely used frontier model at the time of writing, replies “*Anime Characters*” and offers no acknowledgement that the relevant region is unreadable.



(a) Original figure.



(b) One outer label selectively blurred.

Figure 1.2: A sunburst from an arXiv paper in the SCIFIG-EVAL corpus. In **(a)**, the outer-ring category between *Movie Characters* and *Novel Characters* is legibly *Teleplay Characters*; in **(b)**, that single label has been blurred. Asked which category sits between the two, GPT-5.2 answers “*Anime Characters*,” with no admission that the relevant region is unreadable. Source image reproduced from arXiv:2505.23923v1 under its open-access licence; blurred variant generated by the SCIFIG-EVAL pipeline.

Beyond cataloguing such gaps in today’s VLMs, it is essential to understand their behavioural dynamics under a range of visual stressors and, more importantly, to provide a standardised way of reading those dynamics from their outputs, so that the resulting picture can be used to improve the models’ capacity to act as reliable allies in everyday visual work.

1.3 Motivation for Focusing on Scientific Research Figures

Among the domains in which collaborative intelligence is now being tested at scale, scientific research has moved fastest. Between late 2024 and early 2026 the three largest model developers each shipped an autonomous research agent (Google, 2024; OpenAI, 2025b; Anthropic, 2025), and scientific deployments such as the Biomni biomedical platform now report research-cycle compressions of an order of magnitude or more (Anthropic, 2026b). Scientific argument is also overwhelmingly visual: every competent empirical paper eventually commits its claim to a figure, so any system that hopes to act as a credible research partner must read those figures with the fidelity its human collaborator does. The scientific figure is therefore the natural object at which to test whether the current frontier reads visually or only appears to.

Science, in addition, is not a monolingual record. A substantial share of the literature is produced in Chinese, German, Bulgarian and other languages of research, and the cost of excluding that share, in citation, uptake and participation, has been documented at length (Amano et al., 2023). The few multilingual figure- understanding evaluations now appearing already report substantial cross-lingual degradation on tasks the same models solve comfortably in English (Li et al., 2025; Gao et al., 2025; Sharma et al., 2025), which is to say that a figure-reader reliable only in English is, as a research collaborator, structurally partial. The scientific figure considered across languages, then, is what the present work takes as its test object.

1.4 Research Questions

The central objective of this thesis is to characterise, across models, languages and adversarial conditions, how faithfully today's frontier multimodal models actually read scientific figures, and to build an evaluation framework that surfaces the failure modes which conventional accuracy benchmarks obscure. Concretely, the investigation is organised around the following four research questions, each pairing a capacity with the conditions under which that capacity is tested:

RQ1 (Faithful figure description across languages). To what extent do current vision–language models produce accurate, complete and clearly expressed descriptions of scientific figures across typologically diverse languages?

RQ2 (Robustness to visual degradation). How does visual degradation, including noise, compression, aspect distortion and contextual embedding within paper pages, affect the accuracy (whether what is stated is correct), completeness (whether what is visible is reported) and clarity (whether what is reported is expressed unambiguously) of VLM-generated figure descriptions?

RQ3 (Comprehension beyond description). Beyond surface description, can VLMs demonstrate genuine comprehension of scientific visualisations, from answering quantitative questions to performing non-trivial inductive reasoning that requires distinguishing analytical insight from surface-level pattern matching?

RQ4 (Behavioural dynamics under stress). How do VLMs behave when the visual evidence or its accompanying context is stressed, for example through misleading captions, contradictory premises, sycophancy probes, or images whose answer is no longer pixel-recoverable? In particular, do they hold a faithful reading of the figure, adopt the position the prompt implies, or acknowledge that the available evidence does not support a confident answer?

1.5 Research Contributions

In addressing these questions, the thesis makes three contributions.

C1: SCIFIG-EVAL, a multilingual benchmark for scientific figure understanding. We release a corpus of 1,005 figures from 158 arXiv papers, paired with 1,411 expert annotations in English, Bulgarian, German and Chinese, and augmented by a 45-figure adversarial subset covering seven image transforms. Figure understanding is scored along three MQM-adapted axes — accuracy, completeness and clarity — through a dual-judge pipeline designed to control for known judge-model biases.

C2: A psychology-informed adversarial probe design. Grounded in the misinformation effect and the anchoring bias from cognitive psychology, we construct five probe families — hallucination, prompt-reversal, caption-bias, selective-blur and sycophancy — that target distinct stages of the VLM pipeline, from the visual encoder through cross-modal alignment to the autoregressive decoder, and expose failure modes that conventional accuracy benchmarks conflate.

C3: The Admittance–Resistance–Inductance behavioural framework. We propose a three-axis behavioural framework, inspired by the electrical metaphor, that decomposes figure-reading trustworthiness into *Admittance* (honesty in the absence of evidence), *Resistance* (faithfulness under misleading context) and *Inductance* (sound inductive reasoning). Applied across eleven contemporary models from five families, the three axes prove near-orthogonal, recasting trustworthiness as a behavioural profile rather than a single scalar.

1.6 Thesis Structure

The remainder of the thesis is organised as follows. Chapter 2 (Background) provides the technical and conceptual foundation: how frontier multimodal models convert figures into internal representations, how scientific figures carry argument rather than merely depict it, and the failure modes and evaluation traditions on which the remainder draws. Chapter 3 (Literature Survey) situates the work against chart understanding, multimodal hallucination, multilingual evaluation and MQM-style scoring. Chapter 4 (Design) sets out the SciFig corpus, the A–R–L framework and the five adversarial probe families. Chapter 5 (Implementation) describes the evaluation pipeline, the model and judge harness and the reproducibility controls. Chapter 6 (Evaluation) reports the main quantitative results across the 11 evaluated models and the four languages, together with ablations on judge ensembles and probe families. Chapter 7 (Discussion) interprets the findings, connects them to the A–R–L framework and acknowledges the limitations of the study. Chapter 8 (Conclusion) summarises the contributions, revisits the research questions and outlines directions for future work.

Chapter 2

Background

This chapter supplies the technical background required to read the rest of the thesis, organised around four questions addressed in turn by §2.1–§2.4.

2.1 How do today's Vision-Language Models see?

Contemporary vision-language models share a three-stage pipeline. An image is partitioned into patches and mapped to a sequence of embeddings by a vision encoder; the embeddings are projected by a connector into the token space of a language backbone; and the backbone consumes them as a prefix alongside the text prompt (Liu et al., 2023; Alayrac et al., 2022; Li et al., 2023). By the time the image reaches the language model it is a sequence of soft tokens drawn from the same vector space as text, so every downstream operation on the figure is an operation on these tokens.

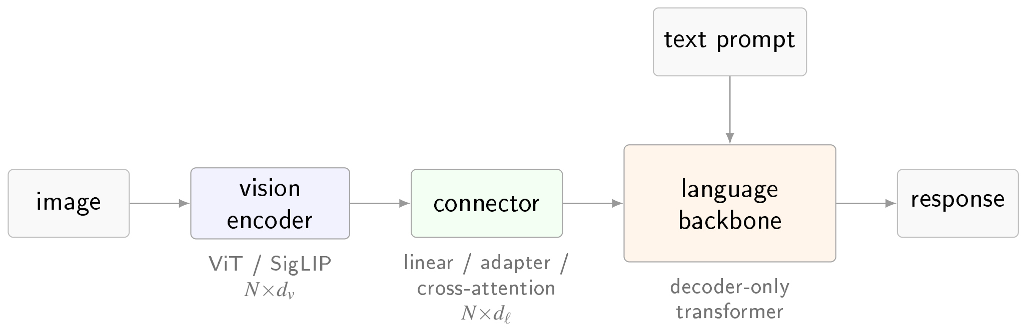


Figure 2.1: The canonical vision-language pipeline. A Vision Transformer (Dosovitskiy et al., 2021), typically trained with a CLIP (Radford et al., 2021) or SigLIP (Zhai et al., 2023) contrastive objective, maps the image to N patch embeddings of width d_v; a connector (linear projection, adapter, or cross-attention module) retunes them to the backbone width d_ℓ (Liu et al., 2023; Li et al., 2023); the decoder-only language model consumes them as a prefix alongside the text prompt.

Four architectural choices distinguish frontier families. The *vision encoder* is most commonly a Vision Transformer (Dosovitskiy et al., 2021) trained with a CLIP (Radford et al., 2021) or SigLIP (Zhai et al., 2023) contrastive objective; a minority, led by Qwen2-VL and Qwen3-VL (Bai et al., 2024; Qwen Team, Alibaba, 2025), train a custom encoder with 2D rotary position embeddings to accept images at arbitrary resolution. The *connector* is either a linear projection into the backbone's embedding space (Liu et al., 2023), a cross-attention adapter with learnable queries of the Q-Former or Perceiver variety (Alayrac et al., 2022; Li et al., 2023), or a set of LoRA modules (Hu et al., 2022) spliced into the backbone as in Phi-4 multimodal (Microsoft Research,

Table 2.1: Publicly documented vision-language architecture across seven representative families spanning the open-weight and closed frontier. "–" denotes a detail that is not disclosed in the primary source. The four open-weight families (top block) publish detailed technical reports; the three closed frontier families (bottom block) publish system cards that address capability and safety but leave the vision pipeline undisclosed. *Sources.* Gemma 3 (Gemma Team, Google DeepMind, 2025); Qwen3-VL (Qwen Team, Alibaba, 2025), with architectural continuity from Qwen2-VL (Bai et al., 2024); Llama 4 (Meta AI, 2025); Phi-4 multimodal (Microsoft Research, 2025); Claude Opus 4.x (Anthropic, 2026a); GPT-5 / 5.2 (OpenAI, 2025a); Gemini 3 Pro (Google DeepMind, 2026).

Family	Vision encoder	Resolution / tiling	Image tokens	Fusion regime
● Gemma 3	frozen SigLIP-400M	896×896 + Pan&Scan	256 (linear proj.)	adapter
● Qwen3-VL	custom ViT + 2D-RoPE	dynamic, any aspect	variable (MRoPE)	native
● Llama 4	MetaCLIP-derived ViT	–	– (early-fusion)	native early-fusion
● Phi-4 mm	CLIP variant + LoRA	–	–	Mixture-of-LoRAs
● Claude Opus 4.x	–	$\leq$ 2576 px long edge	–	–
● GPT-5 / 5.2	–	–	–	native (claimed)
● Gemini 3 Pro	–	up to 900 images / prompt	–	native (sparse MoE)

2025). The *fusion regime* ranges from a late adapter on a frozen backbone, through early fusion on interleaved tokens (Meta AI, 2025; Chameleon Team, 2024), to native co-training of vision and text from initialisation. The *image-token budget*, finally, varies from a fixed allocation per image (Gemma Team, Google DeepMind, 2025) to aspect-preserving variable-length sequences (Bai et al., 2024), and caps the fine-grained detail the encoder can transmit to the backbone. Table 2.1 summarises where seven representative families sit along these axes.

These axes are instantiated very differently by each family. Qwen3-VL (Qwen Team, Alibaba, 2025), shown in Figure 2.2, is illustrative of the open-weight end of the spectrum, with a distinct vision module (preprocessor, vision blocks with interleaved Multimodal Rotary Position Embeddings, and a patch merger) that injects multi-level features into the language backbone through the DeepStack mechanism and supports the dynamic, any-aspect resolution policy in Table 2.1.

Disclosure along these axes is structurally asymmetric. Open-weight families publish technical reports that document the pipeline end to end, whereas closed frontier families publish only system cards (Anthropic, 2026a; OpenAI, 2025a; Google DeepMind, 2026) that leave the encoder, resolution policy, token budget and connector architecture undisclosed. Any evaluation that would attribute an observed output to a specific pipeline stage by reading internal states therefore has, at the frontier, no internal states to read.

2.2 How do today's VLMs reason about what they see, and is it impacted by language?

Reasoning in a vision-language model is autoregressive text generation conditioned on a vision-token prefix. At each decoding step the backbone's self-attention layers attend over a sequence that concatenates projected vision tokens with the text prompt, so every generated token is a function of both channels jointly. Any cognitive operation the model performs on a figure is realised, mechanically, as attention-weighted retrieval from this joint prefix.

Three prompting regimes are layered on this substrate. Under *direct-answer* prompting the

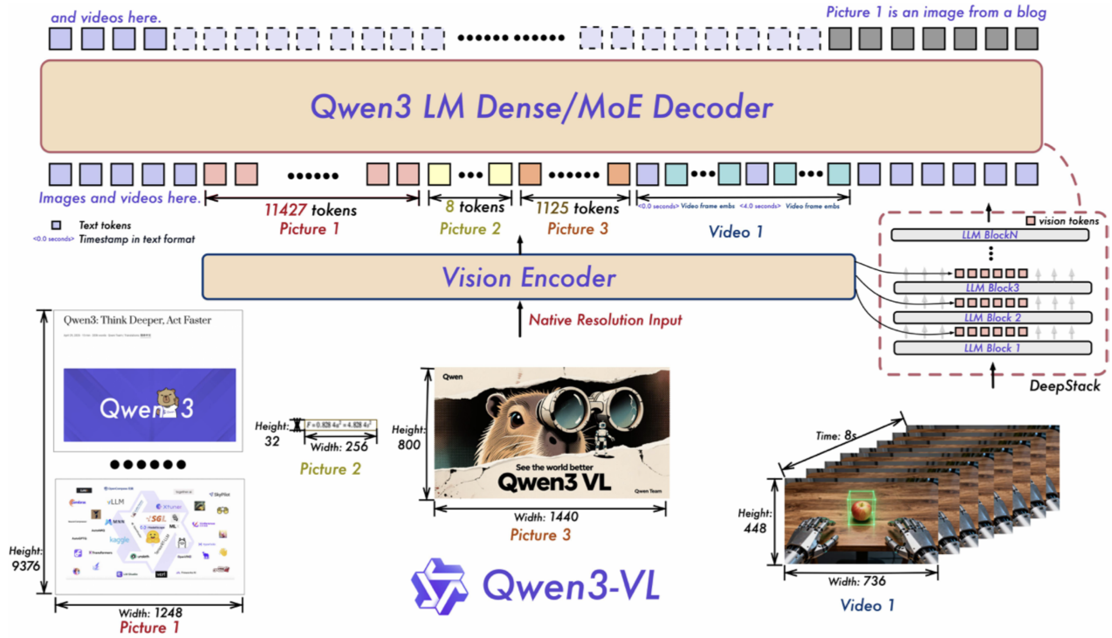


Figure 2.2: Qwen3-VL: a distinct vision module (preprocessor, vision blocks with interleaved M-RoPE, and patch merger) injects multi-level features into the language backbone through the DeepStack mechanism. Reproduced from the Qwen3-VL technical report (Qwen Team, Alibaba, 2025).

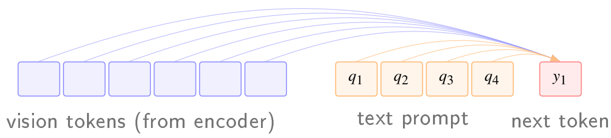


Figure 2.3: At each decoding step the next token is produced by self-attention over a single sequence formed by concatenating the projected vision tokens with the text-prompt tokens; cross-modal reasoning is, mechanically, attention-weighted retrieval over this joint prefix.

model emits its answer immediately, and whatever reasoning occurs is implicit in the transformer computation between the prefix and the first output token. Under *chain-of-thought* (CoT) prompting (Wei et al., 2022) the model first emits a natural-language rationale and then the answer, so that each rationale token re-attends to the vision tokens and spreads visual retrieval across more decoding steps. Multimodal extensions couple this loop to image-space scaffolding. Multimodal-CoT (Zhang et al., 2023) interleaves rationale generation with visual features in a two-stage procedure; Set-of-Mark prompting (Yang et al., 2023) overlays enumerated marks on the image so the rationale can reason over discrete regions by reference; and visual chain-of-thought (Shao et al., 2024) conditions the model to emit region crops during reasoning, effectively re-querying the encoder on sub-regions.

Language enters this pipeline at three structurally distinct layers and perturbs reasoning at each. At the *pixel* layer the vision encoder processes non-Latin scripts through the same fixed patch grid it was pretrained on predominantly for English, so fine-grained symbol recognition (Cyrillic diacritics, Chinese logographs at axis-label size, German compound axis labels) degrades before any language-specific reasoning takes place. At the *token* layer the backbone's tokeniser fragments non-English text into finer subword units, lengthening the sequence over which a claim about the figure is represented and diluting the backbone's lexical priors. At the *prior* layer the pretraining distribution is itself English-dominant, so for an underdetermined chart the conditional distribution over continuations is biased toward English-plausible content. The three layers are largely independent and a given perturbation can act on any one or on all three at once.

2.3 Behavioural Dynamics of VLMs under Challenging or Adversarial Vision Conditions

The pipeline above does not degrade uniformly when an image is ambiguous, partially obscured, or placed in tension with its prompt. Frontier families differ substantially in how their outputs change under stress, and the range of observed dispositions is wider than a single accuracy number suggests. A model may hold a veridical reading of the figure when a leading caption pulls against it, or quietly revise its description to match the asserted position; this latter pattern is the visual analogue of *sycophancy* (Sharma et al., 2023), the tendency of a language model to adopt an interlocutor's implied position when that position is asserted in the prompt. A model may also combine evidence in the image with domain convention and quantitative reasoning to reach a conclusion the image does not literally state, a capacity often unlocked only under chain-of-thought decoding or the image-space scaffolding of §2.2. And when the pixels required to answer are blurred, occluded, or absent, a model may acknowledge the gap and qualify its answer, or produce a confident substituted answer whose tone is indistinguishable from its answers on unperturbed inputs. The text-only counterpart of that last disposition is the calibration behaviour of Kadavath et al. (2022), who showed that stated confidence in a language model can, under suitable elicitation, be made to track accuracy; a matching property on an adequately perceived image is not guaranteed by the architecture and need not hold when the perceptual prefix is deliberately impoverished.

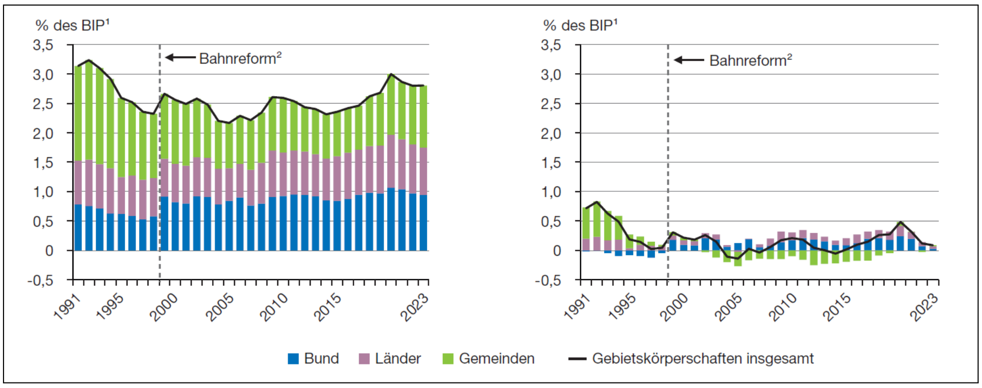
A concrete example makes the phenomenon palpable. Figure 2.4 shows a published German bar-and-line chart of public investment as a percentage of GDP, alongside a variant in which the y-axis ticks, units and legend on the left panel have been blurred while every other visual element, including the right panel's intact axis, remains untouched.

What the empirical literature does not yet supply is a structured way of reading these dispositions off a model's outputs across a controlled stress surface, and that is the gap addressed by the behavioural framework developed in Chapter 4.

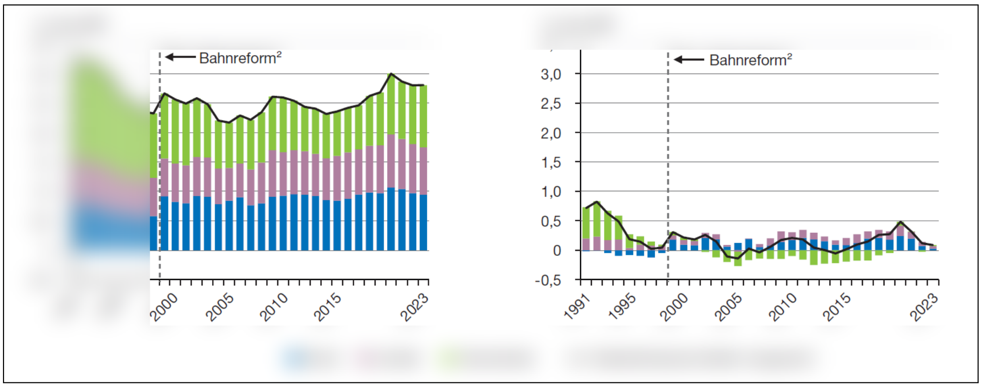
2.4 Evaluating VLM Vision Proficiencies

Open-ended visual evaluation of a VLM places two demands on the measurement apparatus: a rubric more expressive than binary accuracy, since the capabilities indexed in §2.3 are not correct-or-incorrect; and a grading procedure that avoids a human annotator for every candidate, since modern multilingual and perturbation-based evaluations routinely yield tens of thousands of outputs.

The reference rubric is Multidimensional Quality Metrics (MQM), introduced by Lommel et al. (2014) for machine translation and since extended across natural-language generation. Each error is tagged with a category from a fixed typology (accuracy, fluency, terminology, style) and a severity weight (minor, major, critical), and a scalar quality score is computed as a weighted



(a) Original figure as published.



(b) Same figure with the left panel's axis labels, units and legend selectively blurred.

Figure 2.4: An axis-blur stress applied to a published German chart of public investment, 1991–2023. **(a)** the original figure, with axis units (*% des BIP*), tick values and legend (*Bund*, *Länder*, *Gemeinden*) all legible; **(b)** the same figure with those elements blurred on the left panel only. The image alone says nothing about how a model should respond; a faithful reader will acknowledge that the left axis is unreadable, an inferential reader may reconstruct it from the intact right panel and the visible gridlines, and a confidently filling reader will invent units and labels that fit the surrounding text. Source: Werding et al. (2025); blurred variant generated by the SCIFIG-EVAL pipeline.

aggregation of the tagged errors. The apparatus transfers to figure description, with the typology becoming visual fidelity, completeness and clarity and the severity weighting distinguishing a misread tick from a fabricated series.

Human MQM annotation does not scale, and the standard alternative is the *LLM-as-judge* paradigm (Zheng et al., 2023), in which a strong general-purpose model grades candidates against a rubric. The paradigm is reproducible and, under careful deployment, reaches human-level agreement on several tasks, but exhibits documented systematic biases: self-preference toward a judge's own family style (Wataoka et al., 2024); position bias that persists across most judges (Zheng et al., 2023); and length and verbosity biases that reward elaboration largely independent of correctness. Randomising candidate position, averaging across position swaps, and aggregating across a heterogeneous judge ensemble form the methodological baseline for open-ended VLM evaluation.

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